Fast kinematics modeling for conjunction with lens image modeling

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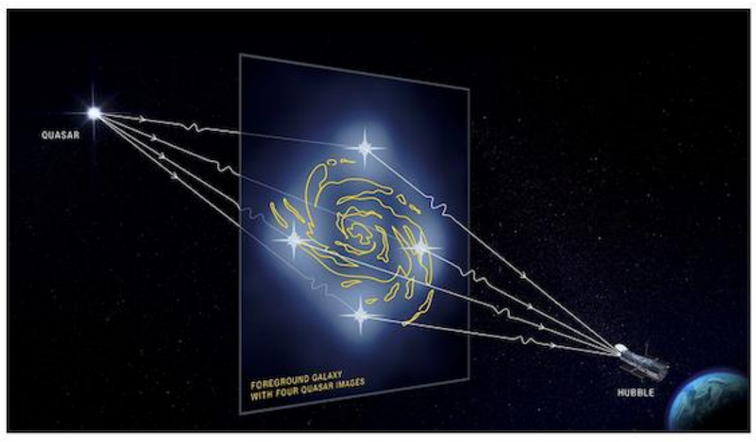
What is gravitational lensing?

Light from a distant source galaxy is deflected by the mass of what is called a **lens galaxy** located between the source and the observer. The observer sees multiple distorted images of the same source. A measure of the time delay between arriving images provides a measure of distance, which constrains **H_0, the current expansion rate of the Universe**. This measure is only accurate if the lens mass model is accurate.

RXJ1131 – 1231;
Image credit: ESA/Hubble, NASA, Suyu et al.



Schematic of gravitational lensing; Image credit: NASA JPL

Kinematics

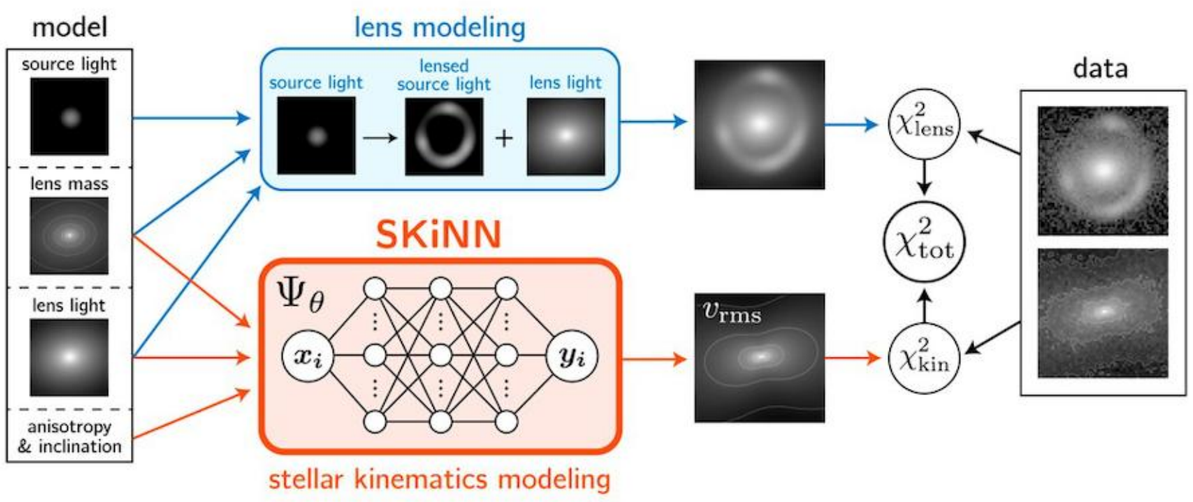
Lensing information alone cannot uniquely constrain the mass model of the lens galaxy; **external information is required**. Galaxy kinematics can provide an independent measure of mass. This requires a separate spectroscopic measurement of the lens galaxy, which gives the **velocity dispersion**. The mass model of the lens must accomodate this velocity dispersion measurement in addition to the lensing information. This constraint breaks the non-uniqueness of lens modeling and recovers a precise H_0.

Until recently, most velocity dispersion measurements have been single-slit aperture measurements. Spherical Jeans kinematic models have been sufficient. Telescopes like JWST are capable of spatially-resolved velocity measurements. This requires **more sophisticated kinematic modeling.**

Current bottleneck

More sophisticated kinematic modeling software exists: Jeans Anisotropic MGE (**JAM**) modeling can calculate the kinematics of ellipsiodal galaxies and would be preferable to spherical assumptions. However, **JAM is slow**; requiring ~15sec per call. In a joint lensing+kinematic modeling procedure, this would have to be called $O(10^5 - 10^6)$ times.This computational bottleneck makes such a joint modeling procedure unfeasible.

Introducing the Stellar Kinematics Neural Network (SKiNN)



Schematic representation of SKiNN's role in a joint modeling framework

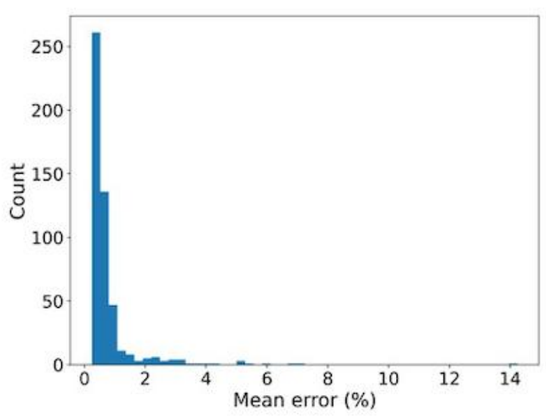
Our insight is to **replace JAM with a NN**, which we name the Stellar Kinematics Neural Network (SKiNN). SKiNN emulates the JAM modeling procedure, but can do so approximately 300 times faster. This speed increase makes it possible to create a **joint framework** which can model lensing and kinematic information together, breaking lensing degeneracies by allowing a full exploration of the joint parameter space.

SKiNN is trained using JAM. The input is a vector of physical parameters describing the lens and the output is a 551x551 image of the velocity (**v_{rms}**) in the plane of the sky. SKINN's architecture consists of 5 blocks each comprising two 2-dimensional convolutional layers followed by an upsampling layer and a ReLU nonlinearity, trained using a standard MSE loss function with the Adam optimizer. For this proof of concept, we have created 5000 v_{rms} images using JAM as a training set (4000 images), test set (500 images), and validation set (500 images). Training SKiNN takes 1 day on 5 Tesla P100 GPUs.

Speed increase

Creating v_{rms} images using JAM takes about 15 seconds per image. SKiNN takes approximately 50ms to call to generate a similar v_{rms} image. We therefore find that, after the upfront training cost, **we speed up the kinematic modeling procedure by roughly a factor of 300.** In practice, this means that the kinematic procedure is no longer the main bottleneck in a joint setting, and is now comparable to the lens modeling procedure, making such a joint framework feasible.

Emulator accuracy



Relative error across the validation set evaluated as the mean error for each image. The 90th percentile is approximately 1%.

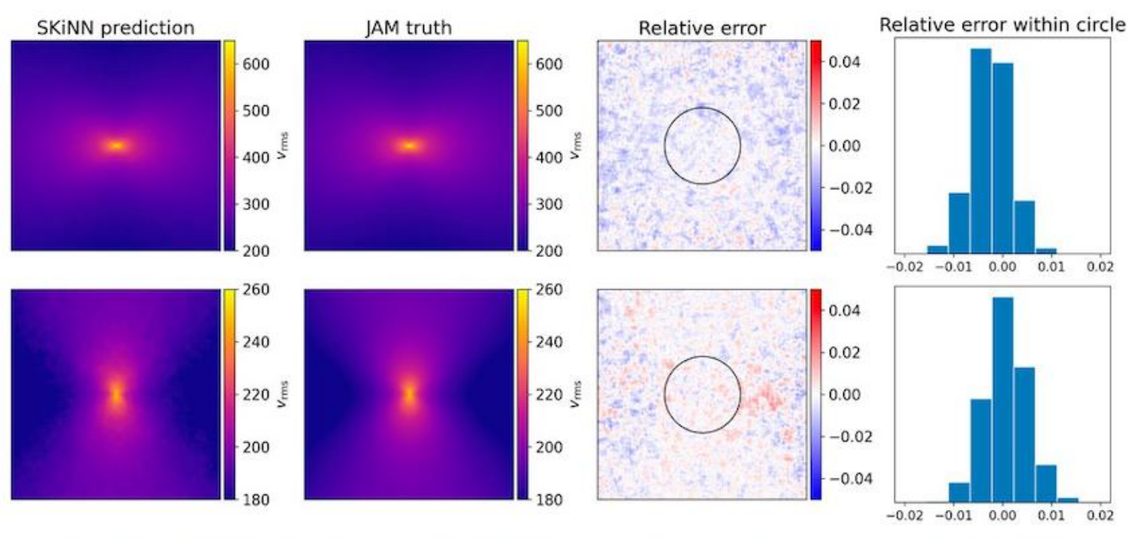
Across the validation set, the error averaged over an image has a median of 0.47%, although some pixels in some images can have error ~few percent. Within the central region (where most of the real observation data points are located), the error is less than 1%. This is lower than the typical systematic uncertainty of $\gtrsim$ 2% in real observed velocity measurements

What's next?

Joint framework: With a working prototype, the next step is currently underway to implement the joint framework in existing lensing software (namely *lenstronomy* and *GLEE*). A coming paper will demonstrate the joint framework on mock systems.

Architecture improvements: Intial experiments appear promising using Conditionally independent Pixel Synthesis (CIPS) generator methods. We are also considering ways to exploit image symmetry,

Generalization: A near-future goal is to build and train SKiNN using more general lensing and kinematics models, now that the method has been demonstrated



Two examble images produced by SKiNN compared to the JAM truth. Residuals within the black circle correspond to the region constrained by real data, with error shown as <1% by the histograms

See paper for References